JOSHUA E. COFFIE
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ATLAS Institute, University of Colorado Boulder

RESEARCH INTERESTS

Bioastronautics, exploration medical systems, human–automation integration, physiological sensing, and advanced materials. Research focuses on the systems-engineering foundations and enabling technologies for progressively Earth-independent medical operations in isolated, confined, and extreme (ICE) environments.

EDUCATION

Ph.D. Student, ATLAS Institute, University of Colorado Boulder | Expected Dec 2027
Research Focus: Design foundations for Earth-independent medical operations, integrating medical systems architecture, physiological sensing, diagnostic decision support, and supervisory intervention.

M.S. in Creative Technology & Design, Social Impact, University of Colorado Boulder | 2023

B.S. in Environmental Geography, U.S. Military Academy at West Point | 2009
Concentration: Environmental Engineering.

TECHNICAL AND RESEARCH COMPETENCIES

- *Systems Engineering*: Functional architecture, requirements analysis, medical capability evaluation, human–system function allocation.
- *Human–Systems Research:* Mixed-methods research, usability evaluation, cognitive workload assessment, trust and reliance, human–automation integration.
- *Bio-HCI & Materials:* Conductive intradermal interfaces, nanomaterial synthesis and characterization, bioelectronic integration.
- *Embedded Systems & Computing:* Arduino, embedded hardware development, sensor integration, rapid prototyping, AI/LLM systems.

RESEARCH & LABORATORY EXPERIENCE

Graduate Researcher | Bioastronautics, Aerospace Engineering Sciences / ATLAS Institute | 2026–Present
- Developed the Sense–Decide–Intervene (SDI) functional framework and an event-driven temporal feasibility criterion to characterize medical-response requirements and the allocation of functions among crewmembers, onboard systems, and terrestrial support.
- Developing an evidence-based methodology for evaluating physiological sensing technologies against event-specific medical information requirements, with applications to persistent intradermal bioelectrical interfaces.
- Conducting a retrospective, dual-cohort mixed-methods evaluation of diagnostic decision aids deployed during 2026 Mars Desert Research Station (MDRS) medical simulations, examining cognitive workload, usability, trust, reliance, and human–system design requirements.
- Developing a theoretical framework for bounded supervisory intervention, examining the allocation of action authority between human operators and automated medical systems.

Graduate Researcher | Emergent Nanomaterial Lab, ATLAS Institute | 2023–Present
- Lead research on implantable conductive tattoo inks for physiological monitoring and Bio-HCI.
- Synthesized and characterized nanoencapsulated ultraviolet photoprotectants and UV-sensitive pigment systems for dermal integration and Bio-HCI applications.

PUBLICATIONS & EXHIBITS

MANUSCRIPTS IN PREPARATION/SUBMITTED

Coffie, J. E., et al. Sense–Decide–Intervene: An Event-Driven Functional Framework for Progressively Earth-Independent Medical Response. Manuscript in preparation, 2026.

Coffie, J. E., et al. Operationalizing SENSE within the Sense–Decide–Intervene Framework: Application to Persistent Physiological Monitoring for Exploration Medical Operations. Manuscript in preparation, 2026.

MANUSCRIPTS SUBMITTED FOR PUBLICATION
Acosta, L., Baker, K., Barsky-Ex, I., Dent, L., LaJoie, E., Coffie, J. E., et al. Insurance and Clinical Scenario in Urogynecology Appointment Access. Urogynecology. Manuscript submitted for publication, August 2026.

CONFERENCE ABSTRACTS SUBMITTED
Coffie, J. E., et al. An Intradermally Implantable Conductive Tattoo for Human–Computer Interaction and Bioelectronic Integration. Abstract submitted to NASA Human Research Program Investigators' Workshop, 2027.

Coffie, J. E., et al. Design and Lessons Learned in Evaluating a Medical Diagnostic Decision Aid Prototype During a Simulated Medical Event in a Surface Analog. Abstract submitted to NASA Human Research Program Investigators' Workshop, 2027.

PATENTS AND EXHIBITS
Joshua Coffie, et al. *"Ink-arnation: Engineering the Human Canvas."* Matter over Mind Exhibit, Dairy Arts Center, 2025.

Joshua Coffie, et al. *"Synthesis and characterization of implantable ultraviolet photoprotectants through nanoencapsulation of bemotrizinol."* Proceedings of ACS Fall 2024.

Fiona Bell, Joshua Coffie, Mirela Alistar. *"Bio-Digital Calendar: Attuning to Nonhuman Temporalities for Multispecies Understanding."* TEI '24.

Fiona Bell, Joshua Coffie, et al. *"μMe: Exploring the Human Microbiome as an Intimate Material for Living Interfaces." DIS '23.*

SELECTED PROJECTS
DeadReckoning Medical Aide | Embedded Diagnostic Decision Aid | MDRS 2026
- Designed, engineered, and field-tested a standalone, offline medical diagnostic decision aid for simulated exploration medical operations at the Mars Desert Research Station.
- Implemented probabilistic differential-diagnosis support and condition-specific medical guidance covering 31 wilderness medical conditions using Arduino-based hardware, an e-paper interface, and a ruggedized enclosure.

Ohm-Sweet-Ohm: Implantable Conductive Ink
- Developed electrically conductive, modifiable intradermal ink formulations for bioelectronic interfacing, supported by a U.S. provisional patent application.
- Investigating applications for persistent physiological sensing, post-implantation electrical modification, and long-duration Bio-HCI interfaces.

SADIE (Scientific Assistant for Data Integration & Experimentation)
- Developed an on-premise, multi-LLM AI system for lab management and grant proposal synthesis.

- Configured for local AI inference with computational offloading to CU Boulder's ALPINE high-performance computing resources.

TEACHING & MENTORING
Guest Lecturer | College of Engineering and Applied Science, CU Boulder | GEEN 1400 | Fall 2026
Teaching Assistant | ATLAS Institute, CU Boulder | 2023
Teacher | Construction, Engineering, & Architecture, BVSD Technical School | 2022–2023
Instructor | Central Texas College (Kandahar Extension, Afghanistan) | 2012–2013

MILITARY & PROFESSIONAL EXPERIENCE
Signal Officer (Captain, O-3) | U.S. Army | 2009–2013
- Planned and managed communications systems, technical operations, and personnel in deployed and resource-constrained environments.
- Served in operational leadership roles, including Aide-de-Camp and Executive Officer assignments within the 10th Mountain Division.

President/Owner | Vigeo Construction and Consulting | 2019–2022
- Managed $3M annual revenue startup focusing on architectural engineering and remodeling.

Client Service Manager | Long View Systems | 2017–2019
- Directed and managed cloud and IT security practices for Denver branch operations.

Senior Consultant | Credera | 2015-2017
Led business and technology consulting projects involving customer experience, user behavior analysis, product development, and enterprise systems integration.

Senior Consultant | KPMG | 2013-2015
Conducted enterprise systems analysis, mergers and acquisitions integration, user experience research, and business process evaluation in support of technology integration, organizational operations, and client engagements.

ACCOLADES & CERTIFICATIONS
- Best Conference Pictorial, TEI ’24
- Bronze Star Medal & Combat Action Badge, U.S. Army (Afghanistan)
- CU Boulder Futures Fellowship, 2024
- WELL AP & LEED Green Associate
- American Sailing Association Certifications | ASA 101, 103, 104, 105, 114